

# CLOUD And BIG DATA SOFTWARE Join Hands With Engineering

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## UMHDL (design and simulation)

An education tool by nature, this software is aimed at making you an expert in learning hardware description language (HDL). Its integrated development environment is intended for learning digital designing with programmable logic devices through simulation. To make it more interactive and easily understandable, it invokes an external VHDL compiler and simulator (such as GHDL), and displays the result of the simulation graphically as waveforms (invoking GTKWave). This application runs on all major platforms including Windows, Linux and Mac OS X.

## LibreCAD (design and simulation)

From drawing a point, line or polyline on your computer screen to working on co-ordinates and layers of your design, LibreCAD does it all. This software is an open source 2D computer-aided design (CAD) application. It is based on Qt, which is a leading cross-platform application and user interface development framework. LibreCAD has layers, blocks, splines, polylines, ellipse tools, advanced tangent lines, circle tools, transformation tools and an advanced snapping system for creating complex designs of tools and machine parts. It is one of those few lightning fast and lightweight CAD software that make use of 2D geometry.

## KiCad (PCB)

KiCad handles all your PCB designing complexities and helps create production-ready electronic schematic diagrams and PCB artwork in no time. KiCad is an open source software tool that can handle up to 32 copper

## Some popular resources

**Amanda.** If you are looking for backup software to back up your data, especially on the server, your search ends here. This stable and free version of software runs on almost all platforms of Linux, Solaris and Windows. Amanda helps back up and archive data from many computers in a network to disk drive or Cloud storage. Written in C and Perl, this software also maintains a catalogue of files that are being backed up from their location on the media.

**ClamAV.** This antivirus software comes from a family of free, cross-platform and open source toolkits that are programmed to detect malicious software and viruses. It could serve as a server-side email virus scanner, too. The application was primarily developed for Unix and has third-party versions available for AIX, BSD, HP-UX, Linux, OS X, OpenVMS, OSF (Tru64) and Solaris. This version is coded to also run on Microsoft Windows platform. Regular updates help keep malicious emails and malwares at bay.

layers, 14 technical layers and four auxiliary layers. It additionally creates all files necessary for building printed boards, Gerber files for photo-plotters, drilling files and component location files. This software also packs a 3D interface for inspecting designs on an interactive canvas. Designers can rotate and pan around so you can inspect details that are difficult to inspect on a 2D view.

Multiple rendering options allow modifications to the appearance of the board, too. KiCad does not present any board size limitation, and its open source capabilities make it the ideal tool for projects oriented towards the creation of electronic hardware, and successful development and maintenance of complex electronic boards.

## ScadaBR (SCADA)

SCADA, or supervisory control and data acquisition, are dedicated software application programs for process control and real-time data gathering from remote locations.

ScadaBR is a Java based platform that offers visualisation of variables, graphs, statistics, protocol configurations, alarms, construction of HMI screens and a number of configuration options. If an unlikely situation is encountered at the factory shop-floor, it sends real-time event and alarm notifications to notify the concerned engineers, who can look for solutions.

## Apache Ambari (utility)

Imagine a complex automated manufacturing factory. It would be ridiculous to have no Cloud based system to monitor and gather data from all machines that are communicating. Apache Ambari project could help monitor clusters of servers that feed into machine data. Ambari makes Hadoop management simpler by providing a consistent, secure platform for operational control. Its Web user interface helps in automating cluster operations while providing centralised security features and additional capabilities using Kerberos and Apache rangers for advanced security. This open source software helps Big Data software fit right into the enterprise environment.

## Alluxio (utility)

With engineering industries relying heavily on data-driven platforms for increasing efficiency and outputs, Alluxio performs best in a distributed environment for Big Data workloads. Engineers who work on Big Data software can utilise its full data-management capabilities by deploying this in existing server clusters through the installation file provided in the DVD accompanying this month's **EFY Plus**. **EFY**

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